Practical 5

20% contribution to the final mark

3 questions, each has 5 points; in total $3 \times 5 = 15$ points

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Assignments

Q1 : Verify (1.1) and (1.2).

Q2 : Consider the model $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \epsilon_i$ for $i = 1, \cdots, n$ and with the $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$. Suppose the parameter $\boldsymbol{\beta}$ is estimated by the ridge regression estimator as in (3.5).

a) The vector of "ridge residuals", defined as $\boldsymbol{\epsilon}(\lambda) = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{\text{ridge}}(\lambda)$, are normally distributed. Why?

b) Show that $\mathrm{E}\boldsymbol{\epsilon}(\lambda) = \boldsymbol{A}\boldsymbol{X}\boldsymbol{\beta}$, where $\left[\boldsymbol{I}_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1}\boldsymbol{X}^\top\right]$.

c) Show that $\mathrm{Var}\,\boldsymbol{\epsilon}(\lambda) = \boldsymbol{A}^\top\boldsymbol{A}\sigma^2$.

d) Could the normal probability plot, i.e. a qq-plot with the quantiles of standard normal distribution plotted against those of the ridge residuals, be used to assess the normality of the latter? Motivate.

Q3 : Consider the following penalized least squares criterion

$$PS_\lambda(\boldsymbol{\beta}) = \|y - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} \hat{w}_i |\beta_j|, \quad \hat{w}_j = \frac{1}{|\hat{\beta}_j|},$$

where $\hat{\beta}_j$ is the jth element of the OLS estimator. For the case of orthonormal design, find the jth component of $\arg\min_{\boldsymbol{\beta}} PS_\lambda(\boldsymbol{\beta})$.